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# The Geological Soma: Seismic Propagation and Tectonic Criticality

[T]-Theory Volume: Geophysics and Earth Sciences

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Alistair Johnson

Independent Researcher, Zurich, Switzerland

ORCID: [0009-0007-2194-0850](https://orcid.org/0009-0007-2194-0850)

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## Introduction: The Earth Has a Somatic Field Too

Seismology has a problem that it rarely states in these terms: it has extraordinarily precise instruments (modern broadband seismometers can detect ground motion of  $10^{-12}$  metres), highly sophisticated computational methods (full-waveform inversion, normal mode summation, finite-difference modelling), and a rich empirical dataset accumulated over a century of observation — but its predictive power for the most consequential events, earthquake nucleation, remains frustratingly limited. The Gutenberg-Richter law tells you the statistical distribution of earthquake magnitudes; it does not tell you when or where the next large event will occur. The Omori law tells you the decay rate of aftershock sequences; it does not tell you which aftershocks will themselves nucleate major events. These are empirical regularities without theoretical derivation.

This book presents a framework that derives those empirical regularities from first principles and provides a mechanism — the WKB tunnelling prediction — for estimating the conditions under which earthquake nucleation will occur. The framework is the **Universal Somatic Field (USF)**, and the key insight is that seismic wave propagation, tectonic criticality, and the rock-mass memory of stress history are all manifestations of the same master field equation that also governs neural dynamics and inter-personal synchronisation.

### Seismic Propagation as a Green's Function

The central claim of this volume is that the seismic P-wave and S-wave propagators are limiting cases of the USF Green's function. The USF Green's function takes the form:

$$G_{\mu\nu}(x, x'; \omega) = \frac{A_{\mu\nu}(\hat{k})}{v_s^2(\mathbf{x}) k^2 - \omega^2 + i\epsilon}$$

where  $v_s(\mathbf{x})$  is the local wave speed (a function of the elastic properties of the medium),  $\hat{k}$  is the propagation direction,  $A_{\mu\nu}$  is the polarisation tensor, and  $\epsilon$  is a small positive damping parameter. This is the standard seismic propagator — the free-space Green's function for the elastic wave equation in a heterogeneous medium. The identification is not approximate; it is exact in the limit where the somatic coupling constants are set to their geological values (zero biological coupling, finite elastic coupling).

This identification has immediate practical implications. Any technique developed for the USF framework applies directly to seismic propagation. In particular, the WKB approximation for barrier tunnelling in the USF framework translates directly into the WKB approximation for seismic wave propagation through slowly-varying media — and then beyond standard WKB, into the nonlinear regime where fault zone physics becomes relevant.

## Tectonic Criticality as Phase Transition

The Gutenberg-Richter law — the power-law distribution of earthquake magnitudes — has been recognised as a signature of criticality since at least the work of Bak and collaborators in the 1990s. The standard account invokes self-organised criticality: the Earth’s crust self-organises to a critical state, and the power-law statistics are the signature of that criticality. This is correct as far as it goes; what it lacks is a derivation of the critical exponent from the physics of the fault system.

In the USF framework, the critical exponent is determined by the universality class of the somatic field phase transition. The somatic field’s energy landscape has a critical point at  $T_c$  where the spectral gap closes and the field dynamics become scale-free. For geological media, the coupling constants are such that the crust operates close to this critical point under normal tectonic loading. The Gutenberg-Richter exponent — typically  $b \approx 1$  — follows from the universality class of the transition, which is determined by the symmetry group of the somatic tensor field.

This is a derivation, not a fit. The framework predicts  $b = 1$  as a consequence of the symmetry group, and deviations from  $b = 1$  in specific geological settings are predicted to correlate with deviations of the coupling constants from the universal values — something that can be tested against regional seismicity data.

## WKB Prediction for Earthquake Nucleation

The most specific and testable prediction of the framework is the WKB estimate for earthquake nucleation. In the USF picture, a fault zone is a system with two stable configurations: locked (stress below the critical shear stress, fault stationary) and slipping (stress above critical, fault in rapid motion). The transition between these configurations — earthquake nucleation — is a barrier-crossing event in the energy landscape. For small stress perturbations, the transition rate is suppressed exponentially; for stress perturbations above the WKB threshold, the transition becomes probable.

The WKB formula gives the transition probability as:

$$P_{\text{nucl}} \approx \exp \left( -\frac{2}{\hbar_{\text{geo}}} \int_{q_1}^{q_2} \sqrt{V(q) - E} dq \right)$$

where  $\hbar_{\text{geo}}$  is the effective geological action quantum (determined by the thermal noise in the fault zone),  $V(q)$  is the potential energy of the fault system as a function of the nucleation coordinate  $q$ , and  $E$  is the current stress energy. This is the standard WKB tunnelling formula, applied to the fault system as a degree of freedom in the somatic field energy landscape.

The practical prediction: the probability of earthquake nucleation grows dramatically as the stress approaches the saddle point of the potential. The rate of growth, and the shape of the potential near the saddle, can in principle be estimated from geodetic measurements of fault locking and stress

accumulation — satellite geodesy, GPS networks, InSAR. The WKB formula then gives a nucleation probability as a function of the measured stress state.

### Rock Strata as Geological Memory

The USF framework gives a natural account of the way fault zones and geological formations encode the history of past stress and deformation. In the Hopfield network picture, the local geological medium stores patterns: fault-parallel fabrics, pressure-solution seams, cataclasite zones all represent previous high-stress configurations that lowered the energy of that configuration in the landscape. A fault that has ruptured before has a memory of that rupture encoded in its geometry, and that memory lowers the energy barrier for future rupture.

This is the formal basis of the empirical observation that faults tend to recur — that major earthquake ruptures tend to follow previous rupture traces. It also provides a quantitative prediction: the memory depth (how far back in geological time the current fault geometry reflects past ruptures) is determined by the decay rate of the stored Hopfield patterns, which is in turn determined by the diffusion rate of the encoding mechanisms (pressure solution, grain growth, mineral recrystallisation).

### The Geographic Connection

The geographic somatic field paper in this volume shows that the same propagator equation governs not just seismic waves but also the large-scale flow of language, culture, and dialect across geographic space. The connection is not superficial: the geodetic topology of the landscape (mountain ranges, river barriers, coastlines) acts on cultural diffusion in exactly the same way that the elastic properties of the crust act on seismic wave propagation. Both are instances of the same heterogeneous-medium Green's function. The USF framework makes this connection mathematically precise.

### What This Book Offers the Geophysicist

The papers assembled here are written for the reader with a background in seismology, tectonics, or Earth sciences. No biology or neuroscience is assumed. The intended reader is comfortable with wave propagation theory, elasticity, and the statistical mechanics of fault systems.

Chapter 2 (geographic somatic field) establishes the propagator identification and the geographic applications. Chapter 3 (zoomable somatic field) develops the scale-invariance that allows the same equation to govern phenomena at geological and sub-geological scales. Chapter 4 (soma-geophysics, the anchor paper written specifically for this volume) develops the seismic propagator identification, the tectonic criticality result, and the WKB nucleation prediction in full detail. The final chapter outlines the research programme: what seismological datasets would test the WKB prediction, and what geodetic measurements would constrain the potential energy function.

The Earth is a field system. The equations have been waiting.

## Introduction

The Universal Somatic Field (Johnson, 2026e) establishes that the Helmholtz Green's function equation:

$$(\nabla^2 + k^2) G(x, x') = \delta(x - x') \quad (1)$$

governs field propagation at twenty scales from quantum foam to the observable universe. Scales 7–9 on the USF dial correspond to animal swarms, human organisms, and societal-scale dynamics. This paper presents worked examples at exactly these scales, drawn from human geography.

The question is not whether the equation applies — that is a theorem (Johnson, 2026e, Section 2) — but what the physical substrate, propagator, and boundary conditions look like at each scale, and whether the predictions match observable patterns. Two examples from the same geographic corridor (the Thames Valley, England) and one from the Swiss Alps are examined.

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## The Thames Valley as a Geographic Wave-Guide

### 2.1 The Substrate

A wave-guide is a physical medium whose boundary conditions preferentially support certain propagation modes and suppress others. A metal microwave wave-guide, an optical fibre, and a submarine canyon are all wave-guides at different scales. The Thames Valley between Heathrow Airport and central London is a geographic wave-guide at Scale 9 ( $10^3$  m).

Its boundary conditions are: - **North wall**: the Chiltern Hills, rising 200–260 m above the valley floor, extending from Oxfordshire to Hertfordshire - **South wall**: the North Downs, rising 150–250 m, extending from Surrey to Kent - **Corridor**: the valley floor along the Thames, 20–40 km wide, containing the highest population density in the United Kingdom outside central London

The Green's function of the corridor selects which propagation modes survive. Patterns with a characteristic wavelength shorter than the corridor width decay within a few kilometres. Patterns with a wavelength matched to the corridor geometry propagate with low loss to the east and west.

### 2.2 Estuary English: A Structural Contagion Wave

Estuary English is a phonological variety characterised by TH-fronting ('th' → 'f'), T-glottalling ('bottle' → 'bo'le'), and L-vocalisation ('milk' → 'miuk'). It has been documented propagating outward from the Thames Estuary since the late twentieth century, moving both geographically along transport corridors and socially upward through the class hierarchy.

On the USF model, this is a **structural contagion wave**: a pattern propagating through a coupled population substrate. Define a population of  $N$  speakers, each with a phonological state  $s_i \in \{0, 1\}$  (0 = RP, 1 = Estuary). The update probability:

$$P(s_i \rightarrow 1) = \sigma \left( \sum_j G_{ij} \cdot s_j - \theta \right) \quad (2)$$

where  $G_{ij}$  is the social interaction kernel (how frequently speakers  $i$  and  $j$  encounter one another),  $\theta$  is a social prestige threshold, and  $\sigma$  is a sigmoid function.

This is a **social Hopfield network** with the Thames Valley Green's function as its propagator. The corridor's geometry selects which phonological modes propagate: variants associated with high-interaction-rate relay nodes (Heathrow, Staines, Richmond, central London) propagate with low loss; variants without these relay stations decay. The documented propagation speed — approximately 30 km per decade along the rail and motorway corridors — is consistent with a diffusion constant set by the interaction frequency at those nodes.

**Equation parameters:**  $k \approx 10^{-3} \text{ m}^{-1}$  (social interaction radius  $\sim 1 \text{ km}$ ); boundary conditions: Chilterns (north), North Downs (south), class prestige gradient (asymmetric coupling in the social dimension);  $N \approx 10^6$  (Greater London population);  $N$  (mind matrix) = cultural attractor count.

### 2.3 Ring-Necked Parakeets: An Active-Matter Velocity Field

The ring-necked parakeet (*Psittacula krameri*) is now the most numerous parrot species in Britain, with a population exceeding 50,000 concentrated in the Thames Valley west of London. Their pre-roost murmurations above the Staines and King George VI reservoirs are large-scale collective phenomena exhibiting the same global coherence as starling murmurations: fluid, topologically connected shapes with no central controller.

On the USF model, this is **Scale 7** (Animal Swarms): the regime where discrete agents dissolve into a continuous active-matter velocity field. The governing equation is the Toner-Tu model:

$$\frac{\partial \mathbf{v}}{\partial t} + \lambda(\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla P + D_T \nabla^2 \mathbf{v} + \eta \hat{\mathbf{n}} \quad (3)$$

where  $\mathbf{v}$  is the local velocity field,  $P$  an effective pressure preventing overlap, and  $\hat{\mathbf{n}}$  the local orientation field. The formation shape emerges from the Green's function of the alignment propagation with the reservoir geometry as boundary conditions.

The Heathrow/Staines reservoir complex acts as a **geographic resonator**: flat water surfaces provide low-turbulence updrafts; surrounding industrial infrastructure supplies the thermal gradients the birds exploit. The roost trajectories are the resonant modes of the active-matter Green's function



evaluated under these boundary conditions. No individual bird stores the formation shape; it is the field's configuration.

**Equation parameters:**  $k \sim r_{\text{align}}^{-1}$  (alignment radius  $\approx 7$  m for parakeets); boundary: reservoir perimeter and surrounding vegetation;  $N$  = flock size ( $\sim 10^4$  at peak roost).

## 2.4 The Same Equation: Two Scales, One Corridor

Both Estuary English (Scale 9,  $10^3$  m) and the parakeet murmuration (Scale 7,  $10^0$ – $10^1$  m) are governed by equation (1) with different wavenumbers and boundary conditions:

| Feature               | Estuary English                              | Parakeet murmuration                      |
|-----------------------|----------------------------------------------|-------------------------------------------|
| Scale                 | 9 (societal)                                 | 7 (animal swarm)                          |
| Characteristic length | $\sim 30$ km propagation                     | $\sim 100$ m formation                    |
| Physical agents       | Individual speakers                          | Individual birds                          |
| State variable        | Phonological variant $s_i$                   | Velocity vector $\mathbf{v}_i$            |
| Propagator            | Social interaction kernel $G_{ij}$           | Alignment force kernel                    |
| Global pattern        | Isogloss wave front                          | Flock formation shape                     |
| Boundary conditions   | Chilterns/North Downs +<br>prestige gradient | Reservoir perimeter + thermal<br>gradient |
| Mind matrix           | Cultural attractors                          | Swarm intelligence<br>(distributed)       |

The Thames Valley selects and amplifies both patterns by the same mechanism: its topographic boundary conditions channel propagation along the east-west axis and suppress transverse modes. Whether the agents are speakers or birds is irrelevant to the propagator equation. Only  $k$  and the physical interpretation of  $G$  change.

*The equation has not changed. Only the substrate has.*

## The Klöntalersee: A Parabolic Acoustic Resonator

The Klöntalersee is a glacially carved lake in Canton Glarus, eastern Switzerland. Its geometry approximates a parabolic bowl — approximately 3 km long, 0.5 km wide — with near-vertical limestone walls rising 1,000 m on the south side (the Glärnisch massif). The north side descends more gradually toward the Glarus valley floor.

At Scale 10 (geological,  $10^5$  m), the lake basin is a **natural acoustic Green's function evaluator**. A parabolic boundary reflects incoming waves and focuses them at the focal point of the parabola. Acoustic measurements in such valleys consistently show anomalously long reverberation times

compared to open terrain — the boundary conditions confine the acoustic field and sustain resonant modes that would otherwise decay.

The Glarus Hauptüberschiebung (Glarus Overthrust), the UNESCO World Heritage geological formation immediately adjacent to the lake, provides the seismic counterpart: 250 Ma Verrucano sandstone resting on 35 Ma Eocene flysch, with 35 km of northward transport recorded. This is a seismic wave with a ten-million-year period — the same Green's function at Scale 10 ( $10^5$  m) with a period of geological time rather than acoustic time.

**Equation parameters:** acoustic:  $k = \omega/c_{\text{air}} \approx 2\pi f/340 \text{ m}^{-1}$  (e.g.,  $f = 100 \text{ Hz}$ :  $k \approx 1.8 \text{ m}^{-1}$ ); seismic:  $k = \omega/v_P$  ( $v_P \approx 6000 \text{ m/s}$ ); boundary: valley walls (limestone, high acoustic impedance);  $N$  (mind matrix) = crustal stress mode count.

Both the acoustic resonator and the seismic record are instantiations of the same Green's function equation with different wavenumber  $k$  and different physical interpretation of “source” and “response.”

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## Discussion

### 4.1 Geographic Boundary Conditions as Scale Selectors

The central insight of this paper is that geographic features function as boundary conditions on the Green's function equation at Scale 7–10. Mountain ranges, valley floors, coastlines, and reservoir complexes select which propagation modes survive long-range transmission. This is not a metaphor; it is the same mathematical mechanism as the boundary conditions of a microwave cavity or an optical fibre.

The implications for cultural geography are direct. The propagation of languages, species ranges, technological adoption curves, and disease vectors all follow patterns consistent with equation (1) evaluated under the boundary conditions of the underlying geographic substrate. The rate and direction of propagation are determined by the Green's function of the landscape, not by the intrinsic properties of the propagating pattern.

### 4.2 Relation to the Universal Somatic Field

This paper adds Scales 7–10 to the USF's empirical base. The existing papers in this collection establish the framework at Scale 5 (cellular/neural), Scale 6 (brain/CEMI), Scale 8 (organism), and Scales 13–20 (stellar to cosmological). The geographic scale (7–10) was the missing middle — the regime where biological agents aggregate into collective phenomena and where physical geography provides the boundary conditions.

The conclusion of the USF framework — that the same equation governs all twenty scales — gains additional support from the examples presented here. The Thames Valley corridor is not a special

case; it is a particularly legible one. The same physics operates in every geographic feature. The parabolic bowl of the Klöntalersee, the Thames Valley wave-guide, and the Himalayan watershed are all Green's function evaluators at different scales, with different  $k$  values and different physical substrates.

### 4.3 Neurodivergent Pattern Recognition

The identification of structural similarity across wildly different scales — parakeet murmurations and dialect spread in the same geographic corridor, governed by the same equation — is an example of the cross-domain pattern recognition that characterises atypical cognitive profiles (ASC Level 2, ADHD) as described in the companion paper on the pre-verbal manifold (Johnson, 2026d).

Neurotypical processing compresses the high-dimensional state of perception into a low-dimensional narrative, discarding cross-domain structural parallels as noise. Less-compressing processing retains these parallels as signal. The connection between a dialect wave and a bird swarm is not obvious to sequential, narrative-linear processing; it is immediate to field-theoretic, parallel processing. This is not a character trait; it is a parameter setting in the FM-HN architecture (Johnson, 2026c).

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## Conclusion

The Thames Valley supports two simultaneous examples of scale-invariant field propagation: Estuary English as a structural contagion wave at Scale 9, and ring-necked parakeet murmurations as an active-matter velocity field at Scale 7. Both are governed by the Green's function of the valley's geographic wave-guide, evaluated at their respective wavenumbers. The Klöntalersee basin provides a third example at Scale 10: a parabolic acoustic resonator whose seismic counterpart records a ten-million-year wave.

In all three cases, the equation is the same. Only the substrate, the wavenumber, and the physical interpretation of source and response differ.

The geographic somatic field is not a new theory. It is the Universal Somatic Field evaluated at geographic boundary conditions. The field is always there. The geography makes it visible.

---

## Introduction

*They saw a guitar string. I heard the music.*

— A.J. (after Feynman)

The search for a unified description of physical reality has proceeded, since Newton, by identifying common mathematical structures across phenomena that appear superficially different. Fourier analysis revealed that the vibration of a string, the propagation of heat, and the conduction of electricity all obey the same equation with different boundary conditions. Maxwell showed that electricity and magnetism are aspects of a single field. Einstein showed that gravity and acceleration are locally indistinguishable.

The present work identifies a further unification: the same Green’s function equation that governs electromagnetic propagation at the atomic scale ( $10^{-10}$  m) also governs seismic propagation at the geological scale ( $10^5$  m), cortical electromagnetic propagation at the neural scale ( $10^{-1}$  m), and gravitational wave propagation at the cosmological scale ( $10^{26}$  m). The wavenumber  $k$  changes at each scale; the equation does not.

This is not the observation that “waves are everywhere” — a qualitative truism — but a precise structural claim: the Green’s function of any substrate with a characteristic oscillation frequency  $k$  satisfies the Helmholtz equation  $(\nabla^2 + k^2)G = \delta$ , and the solutions of this equation have the same form regardless of the physical medium. The propagator  $G$  is the medium’s impulse response: its answer to the question *what happens at  $x$  given a unit perturbation at  $x'$ ?*

This identification has a consequence for string theory. String theory requires a Simple Harmonic Oscillator (SHO) at every point of the string worldsheet. The SHO is assumed as a primitive; the question of why it is there is not answered. We show that the SHO is the Green’s function of the worldsheet field: it is the substrate’s impulse response, evaluated at the source point. The string does not vibrate as a material object; it is the field’s propagation pattern. This is not a reinterpretation — it is a derivation from the structure of field equations.

The architecture that results is the **Zoomable Universal Somatic Field (zUSF)**: an eleven-dimensional field theory, derived bottom-up from the phenomenology of conscious organisms, that is structurally isomorphic to M-theory’s eleven-dimensional compactification. The isomorphism is not metaphorical; it is a type-level proof verified by the Lean 4 kernel (`MTheoryIsomorphism.somaField_iso_mtheory`).

The derivation is inductive rather than deductive. Where Veneziano (1968) wrote down a scattering amplitude and Nambu, Nielsen, and Susskind separately identified the string as its underlying object, the present work identifies the Green’s function as the oscillator. Where M-theory arrived at eleven dimensions from mathematical consistency requirements, the present work arrives at eleven

by counting the functional degrees of freedom of a living organism. That the two derivations agree is the isomorphism at the heart of this paper.

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## Mathematical Foundation

### 2.1 The Master Equation

The foundational equation is the Helmholtz Green's function equation:

$$(\nabla^2 + k^2) G(x, x') = \delta(x - x') \quad (1)$$

where  $x, x' \in \mathbb{R}^3$ ,  $k > 0$  is the wavenumber, and  $\delta$  is the Dirac delta distribution. Equation (1) admits the free-space solution:

$$G(x, x') = \frac{e^{ik|x-x'|}}{4\pi|x-x'|} \quad (2)$$

This is the retarded propagator: the field amplitude at  $x$  due to a unit point source at  $x'$ . Three properties are immediate:

1. **Positivity of amplitude:**  $|G| > 0$  for all  $x \neq x'$ .
2. **Decay:**  $|G| \sim 1/r$  as  $r = |x - x'| \rightarrow \infty$  (radiation condition).
3. **SHO structure:** For fixed  $x$ , the function  $x' \mapsto G(x, x')$  satisfies  $(\partial^2/\partial x'^2 + k^2)G = \delta$  — the harmonic oscillator equation in the source variable.

Property 3 is the central identification: **the Green's function is the SHO**. The SHO of string theory, required at every worldsheet point, is the substrate's impulse response. This observation, formalised in the companion file `UniversalSomaticField.lean` (`axiom greens_fn_is_SHO`), is the structural core of the zUSF.

### 2.2 Scale Invariance

As  $k$  varies from  $k_P = \ell_P^{-1} \approx 10^{35} \text{ m}^{-1}$  (Planck scale) to  $k_H = \ell_H^{-1} \approx 10^{-26} \text{ m}^{-1}$  (Hubble scale), the form of equation (1) is invariant. The boundary conditions and the physical interpretation of  $G$  change; the equation does not.

**Definition (Scale-Invariant Field).** A field  $G$  is *scale-invariant* if for every scale  $\sigma \in \{0, \dots, 20\}$  there exists a wavenumber  $k(\sigma) > 0$  and a physical substrate  $\mathcal{S}(\sigma)$  such that  $G$  satisfies equation (1) with those parameters.

**Theorem (Lean 4 verified, `UniversalSomaticField.universal_field_theory`):**  $G$  is scale-invariant across all 20 levels.

The 20-step scale dial: each level coloured from violet (Planck) to yellow (cosmic). The master equation  $(\nabla^2 + k^2)G = \delta$  is invariant across all levels; only  $k$  changes. (Generated: *FA\_universal\_dial.png*)

Figure 1: The 20-step scale dial: each level coloured from violet (Planck) to yellow (cosmic). The master equation  $(\nabla^2 + k^2)G = \delta$  is invariant across all levels; only  $k$  changes. (Generated: *FA\_universal\_dial.png*)

Correspondence Principle:  $\text{softmax}(+1)$  converges to 1 as  $\beta \rightarrow \infty$  (log scale). At  $\beta = 50$ , the output is numerically indistinguishable from the classical sign function. (Generated: *FSx\_softmax\_correspondence.png*)

Figure 2: Correspondence Principle:  $\text{softmax}(+1)$  converges to 1 as  $\beta \rightarrow \infty$  (log scale). At  $\beta = 50$ , the output is numerically indistinguishable from the classical sign function. (Generated: *FSx\_softmax\_correspondence.png*)

*Proof.* By `scale_invariance_inhabited`: for every  $\sigma$ , the type `FieldEquation`  $\sigma$  is inhabited.  $\square$

### 2.3 Log-Sum-Exp and the Correspondence Limit

For the biological substrate at Scale 6, the propagator takes a modified form. The FM-HN architecture (§7) uses the log-sum-exp energy:

$$E_{20}(\xi) = -\frac{1}{\beta} \log \sum_{\mu} e^{\beta \xi^{\mu T} \xi} + \frac{1}{2} \|\xi\|^2 \quad (3)$$

whose update rule  $\xi \leftarrow X^T \cdot \text{softmax}(\beta \cdot X\xi)$  converges to the classical sign update as  $\beta \rightarrow \infty$ :

$$\lim_{\beta \rightarrow \infty} \text{softmax}(\beta \cdot z)_i = \mathbf{1}[i = \arg \max z] \quad (4)$$

This limit — the Correspondence Principle — is verified in `LimbicHopfield.correspondence_principle` and illustrated in figure 2.

---

## The Eleven-Dimensional Architecture

### 3.1 Decomposition

Let  $\mathcal{M}_{11}$  denote the configuration space of a living organism in interaction with its environment. We decompose  $\mathcal{M}_{11}$  as:

$$\mathcal{M}_{11} = \underbrace{M_4}_{\text{Spacetime}} \times \underbrace{P_3}_{\text{Propagator}} \times \underbrace{L_1}_{\text{Limbic}} \times \underbrace{C_3}_{\text{Cortex}} \quad (5)$$

The four subspaces are:

| Symbol | Dim | Physical substrate                         | Mathematical role                        |
|--------|-----|--------------------------------------------|------------------------------------------|
| $M_4$  | 4   | Body in 3+1D<br>spacetime                  | Lorentzian manifold,<br>causal structure |
| $P_3$  | 3   | Endogenous EMF<br>(CEMI field)             | Green's function<br>domain               |
| $L_1$  | 1   | Limbic axis<br>(homeostatic<br>regulation) | Orbifold segment<br>[−1, 1]              |
| $C_3$  | 3   | Cortical information<br>routing            | Green's function<br>co-domain            |

The compact 7-dimensional internal space is:

$$X_7 = P_3 \times L_1 \times C_3, \quad \dim(X_7) = 3 + 1 + 3 = 7 \quad (6)$$

**Theorem (Lean 4 verified, `MTheoryIsomorphism.dim_is_11`):**  $4 + 3 + 1 + 3 = 11$ . Proof by decide.  $\square$

### 3.2 Isomorphism with M-Theory

M-theory (Witten 1995) compactifies eleven-dimensional supergravity as  $M_{11} = M_4 \times X_7$  where  $X_7$  is a compact manifold with  $G_2$  holonomy. The soma-field decomposition (5) has identical dimensional structure.

**Theorem (Lean 4 verified, `MTheoryIsomorphism.somaField_iso_mtheory`):** There exists a type isomorphism:

$$\text{SomaField}_{11} \cong \text{Spacetime} \times \text{CompactSpace}_7 \quad (7)$$

*Proof.* By `toMTheory` and `fromMTheory`; roundtrip by `simp`.  $\square$

The derivation is independent: the M-theory structure was not assumed; it was arrived at by counting functional degrees of freedom of a biological system. The isomorphism (7) is therefore a theorem, not a construction.

### 3.3 The Limbic Axis as a Horava-Witten Orbifold

In Horava-Witten M-theory (1996), the compact direction is an orbifold  $S^1/\mathbb{Z}_2$  — a line segment with two ten-dimensional boundary spacetimes at each endpoint. The Limbic Axis  $L_1 \cong [-1, 1]$  has the same structure:

- Endpoint  $x = -1$ : somatic boundary (body-world)
- Endpoint  $x = +1$ : cortical boundary (mind-world)

The quartic double-well potential  $V(x) = W(x^2 - 1)^2$  for barrier heights  $W \in \{8, 10, 12\}$  corresponding to the QUANT-EXP-1 sweep. (Generated: *FS0\_double\_well.png*)

Figure 3: The quartic double-well potential  $V(x) = W(x^2 - 1)^2$  for barrier heights  $W \in \{8, 10, 12\}$  corresponding to the QUANT-EXP-1 sweep. (Generated: *FS0\_double\_well.png*)

- Interior  $(-1, 1)$ : transition zone, subject to quantum tunnelling

**Theorem (Lean 4 verified, `MTheoryIsomorphism.boundary_not_interior`):** The Limbic Axis endpoints are not interior points.  $\square$

The double-well potential on  $L_1$ :

$$V(x) = W(x^2 - 1)^2 \quad (8)$$

models the energy barrier between somatic and cortical attractors. At  $x = -1$  (trauma attractor), classical gradient descent is trapped: `LimbicTunnel.gradient_traps_near_neg1` (proved by `nlinarith`).

---

## The Zoom Operator

### 4.1 Definition

**Definition (Zoom Operator).** The Zoom Operator  $\Lambda$  is a dependent type constructor:

$$\Lambda : \text{ScaleLevel} \rightarrow \text{FieldEquation} \quad (9)$$

where  $\text{ScaleLevel} = \text{Fin}(21)$  and  $\text{FieldEquation}(\sigma)$  packages the wavenumber  $k(\sigma)$ , boundary conditions, and substrate type.

In Lean 4:

```
structure FieldEquation (n : ScaleLevel) where
```

```
  k : ℝ
```

```
  hk : 0 < k
```

```
  G : ℝ → ℝ → ℝ
```

**Remark on the choice of 20 levels.** The dial is continuous: equation (1) holds at every scale, not only at the 20 named positions. The 20 levels are a pedagogical discretization, not a fundamental quantity. The choice is motivated by two considerations. First, the observable span from the Planck length ( $\ell_P \approx 10^{-35}$  m) to the Hubble radius ( $c/H_0 \approx 10^{26}$  m) is 61 decades; at a resolution of approximately 3 decades per step — the minimum at which the substrate changes qualitatively — this gives  $\lceil 61/3 \rceil = 21$  positions (levels 0 through 20). Second, the 20 steps coincide with five



major qualitative phase transitions at which the governing physics changes character, each spanning approximately four steps:

| Transition              | Levels | Nature of change                                                            |
|-------------------------|--------|-----------------------------------------------------------------------------|
| Quantum → Classical     | 0–4    | Spacetime geometry emerges;<br>probability amplitude<br>collapses to matter |
| Chemistry → Biology     | 4–7    | Self-replication and<br>homeostatic regulation appear                       |
| Individual → Collective | 7–10   | Agency distributes across<br>coupled agents                                 |
| Geological → Stellar    | 10–14  | Self-gravity dominates over<br>chemical binding                             |
| Stellar → Cosmic        | 14–20  | Dark energy and expansion<br>compete with gravity                           |

The number 20 is therefore not arbitrary, but it is also not uniquely determined. A discretization into 15 or 25 steps would be equally defensible. The scientific claim is about the invariance of equation (1), not about the count of steps. The steps are tick marks on a continuous dial.

## 4.2 Physical and Mind Scaling

Physical scaling proceeds through the characteristic length  $\ell(\sigma) \sim k(\sigma)^{-1}$ , ranging from  $10^{-35}$  m ( $\sigma = 0$ ) to  $10^{26}$  m ( $\sigma = 20$ ).

Mind scaling proceeds through the tensor rank  $N(\sigma)$ :

$$N(\sigma) \approx \begin{cases} 10^2 & \sigma = 2 \text{ (nuclear)} \\ 10^{14} & \sigma = 6 \text{ (brain)} \\ 10^{11} & \sigma = 15 \text{ (galactic)} \\ \infty & \sigma = 20 \text{ (universal)} \end{cases} \quad (10)$$

**Theorem (architectural constraint):** Physical scale  $\ell(\sigma)$  and mind rank  $N(\sigma)$  zoom together. They cannot zoom independently.

*Proof.* The Zoom Operator returns a dependent pair  $(\ell(\sigma), N(\sigma))$  whose components are both functions of the same  $\sigma$ . Setting  $\sigma$  determines both simultaneously.  $\square$

Physical scale (left, log metres) and mind rank  $N$  (right, log units) both increase with  $\sigma$  from 0 to 20. The two bars are tethered — a change in one forces a change in the other. (*Generated: FA\_dual\_scaling.png*)

Figure 4: Physical scale (left, log metres) and mind rank  $N$  (right, log units) both increase with  $\sigma$  from 0 to 20. The two bars are tethered — a change in one forces a change in the other. (*Generated: FA\_dual\_scaling.png*)

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## The Twenty-Scale Catalogue

This section instantiates equation (1) at each of the twenty scale levels. For each level we state: the physical substrate, the Green's function interpretation, the mind matrix, and the equation parameters. The pattern is invariant: only the labels change.

---

### Scale 0 — Quantum Foam ( $10^{-35}$ m)

**Equation parameters:**  $k = k_P = \ell_P^{-1} \approx 10^{35} \text{ m}^{-1}$ ; boundary: periodic (no preferred direction);  $N = \infty$  (all configurations in superposition).

**Physical substrate:** Discrete spacetime nodes; pre-geometric fluctuations. At the Planck scale, geometry itself becomes probabilistic. The metric fluctuates with amplitude  $\delta g \sim 1$  (Planck units).

**Propagator:**  $G$  is the gravitational quantum amplitude — the probability amplitude for a graviton to propagate from  $x'$  to  $x$ . In the semiclassical limit:  $G_P(x, x') = \langle x | \hat{G} | x' \rangle$  (Feynman path integral). The worldsheet SHO is  $G$  evaluated at the string scale (Scale 1).

**Mind matrix:** Quantum superposition state. The S-matrix encodes all possible scattering outcomes as complex amplitudes.  $N_0 = \dim(\mathcal{H}_{\text{Planck}}) = \infty$ .

**Remark.** Scale 0 is where equation (1) takes its most abstract form. Every subsequent scale is this equation, coarse-grained.

---

### Scale 1 — String Scale ( $10^{-32}$ m)

**Equation parameters:**  $k = \ell_s^{-1} \approx 10^{32} \text{ m}^{-1}$ ; boundary: periodic (closed string) or Dirichlet (open string on D-brane).

**Physical substrate:** String worldsheets; M-theory 2-branes. The string characteristic length is  $\ell_s \approx 10^{-32} \text{ m}$ .

**Propagator:** The worldsheet Green's function:  $G_{\text{string}}(\sigma, \sigma') = -\frac{\alpha'}{2} \ln |\sigma - \sigma'|^2$  (free bosonic string, Regge slope  $\alpha'$ ). The vibrational modes satisfy the SHO equation  $\ddot{X}^n + n^2 X^n = 0$ .

Left: the Simple Harmonic Oscillator ( $\ddot{x} + \omega^2 x = 0$ ). Right: the Green's function  $G(\tau)$  of a harmonic system — both satisfy the SHO equation. They are the same object. (Generated: *FS1\_sho\_string.png*)

Figure 5: Left: the Simple Harmonic Oscillator ( $\ddot{x} + \omega^2 x = 0$ ). Right: the Green's function  $G(\tau)$  of a harmonic system — both satisfy the SHO equation. They are the same object. (Generated: *FS1\_sho\_string.png*)

Yukawa potential  $e^{-mr}/r$  (nuclear, solid) versus Coulomb potential  $1/r$  (electromagnetic, dashed). Same master equation; different mass parameter  $k$ . (Generated: *FS2\_yukawa\_vs\_coulomb.png*)

Figure 6: Yukawa potential  $e^{-mr}/r$  (nuclear, solid) versus Coulomb potential  $1/r$  (electromagnetic, dashed). Same master equation; different mass parameter  $k$ . (Generated: *FS2\_yukawa\_vs\_coulomb.png*)

**Key result.** The SHO of string theory IS  $G$ . A string vibrational mode at frequency  $n$  is the  $n$ -th Fourier mode of the worldsheet's impulse response. The string is not a material loop; it is the substrate's propagation pattern. This is `UniversalSomaticField.greens_fn_is_SHO` (axiom).

**Mind matrix:** The string landscape:  $N \sim 10^{500}$  vacuum configurations. Each selects a different low-energy physics. Our universe occupies one vacuum.

---

## Scale 2 – Nuclear ( $10^{-15}$ m)

**Equation parameters:**  $k = m_\pi c/\hbar \approx 10^{15} \text{ m}^{-1}$ ; boundary: confinement radius  $r \lesssim 1 \text{ fm}$ .

**Physical substrate:** Quarks, gluons, atomic nuclei. Strong nuclear force confines quarks; the residual strong force between nucleons is mediated by pion exchange.

**Propagator:** Yukawa kernel:  $G_{\text{nuc}}(r) = e^{-m_\pi r}/(4\pi r)$ . The exponential factor  $e^{-m_\pi r}$  encodes finite range. Setting  $m_\pi = 0$  recovers the Coulomb propagator of Scale 3 — the transition from a massive to a massless carrier.

**Mind matrix:** Nuclear S-matrix ( $N \approx 10^5$  nuclear energy levels across all stable nuclei). Binding energy curve = eigenvalue spectrum of  $G_{\text{nuc}}$ .

---

### Scale 3 – Atomic ( $10^{-10}$ m)

**Equation parameters:**  $k = \sqrt{2m_e E}/\hbar$ ; boundary: molecular orbital extent;  $k = 0$  for the static Coulomb case.

**Physical substrate:** Atoms; electron orbitals; covalent bonds. The characteristic length is the Bohr radius  $a_0 = 0.529$  Å.

**Propagator:** Coulomb kernel:  $G_{\text{EM}}(r) = 1/(4\pi r)$ . This is the  $k \rightarrow 0$  limit of equation (2): the photon is massless, giving infinite range. The Schrödinger equation with Coulomb potential generates atomic orbitals as eigenfunctions of  $G_{\text{EM}}$ .

**Key property.** The first infinite-range propagator in the catalogue. Electromagnetism reaches across the observable universe.

**Mind matrix:** Atomic orbital basis ( $N \approx 10^2$  per atom). Ionisation energy = barrier height of the atomic attractor.

**Same as always.** The  $1/r$  dependence of  $G_{\text{EM}}$  at atomic scale ( $10^{-10}$  m) is identical in form to the gravitational propagator at cosmological scale ( $10^{26}$  m). Equation (1) with  $k = 0$ .

### Scale 4 – Molecular ( $10^{-9}$ m)

**Equation parameters:**  $k = \sqrt{2m_e E_{\text{bond}}}/\hbar$ ; boundary: nuclear positions and molecular geometry.

**Physical substrate:** Chemical bonds; crystal lattices; biological macromolecules (proteins, DNA). Molecular geometry is the ground state of the electron density under nuclear boundary conditions.

**Propagator:** Schrödinger Green's function: electron density amplitude  $G_e(x, x') = e^{ik|x-x'|}/(4\pi|x-x'|)$ . Molecular orbitals are eigenmodes of  $G_e$  — resonances of the propagator under nuclear constraints.

**Mind matrix:** Molecular conformational space ( $N \approx 10^3$  per protein). Protein folding = energy minimisation on the molecular attractor landscape. A conformational transition (e.g., retinal 11-cis  $\rightarrow$  all-trans, triggering vision) is an attractor transition in  $G_e$ .

**Same as always.** The molecular conformation double-well  $V(x) = W_{\text{mol}}(x^2 - 1)^2$  with  $W_{\text{mol}} \approx 4$  eV is identical in structure to the limbic double-well (Scale 6,  $W \approx 10$  natural units) — separated by twenty-five orders of magnitude in characteristic length.

### Scale 5 – Cellular / Neural ( $10^{-6}$ m)

**Equation parameters:**  $k = \lambda_{\text{axon}}^{-1} \approx 2000 \text{ m}^{-1}$  (axon space constant  $\lambda \approx 0.5 \text{ mm}$ );  $N \approx 10^4$  per neuron.

**Physical substrate:** Neurons; synapses; axon fibres; fascial networks. The cable equation  $(\partial^2/\partial x^2 - \lambda^{-2})V = I_{\text{inj}}$  is the hyperbolic form of equation (1) with  $k = i\lambda^{-1}$ .

**Propagator:** Synaptic transfer function. Neural impulse response encodes how a spike at the pre-synaptic terminal propagates to the post-synaptic membrane. Ephaptic coupling (Anastassiou et al. 2011) extends this to the electric field generated by synchronised neural firing.

**Mind matrix:** Synaptic weight matrix  $W_{ij}$  (Hopfield 1982). Stored memories are local minima of  $E_{82}(s) = -\frac{1}{2}s^T W s$ . Storage capacity: approximately  $0.14 \cdot D$  patterns for  $D$ -dimensional state space.

**Key result.** QUANT-EXP-1 (Johnson, 2026a): quantum annealing achieves escape from the trauma attractor (barrier  $W \in \{8, 10, 12\}$ ) with 3/3 success rate; classical Langevin dynamics achieve 0/48. WKB tunnelling amplitude:

$$\Theta(W) = \exp\left(-\frac{8\sqrt{2W}}{3}\right) > 0 \quad \forall W > 0 \quad (11)$$

(proved: `LimbicTunnel.wkbAmplitude_pos`).

---

### Scale 6 – Brain / CEMI Field ( $10^{-1}$ m)

**Equation parameters:**  $k = \omega/c_{\text{neural}} \approx 2\pi \times 40 \text{ Hz}/6 \text{ m s}^{-1} \approx 40 \text{ m}^{-1}$  (gamma band);  $N \approx 10^{14}$  (synaptic connections).

**Physical substrate:** Cerebral cortex; subcortical nuclei; 86 billion neurons; approximately  $10^{14}$  synapses organised into cortical layers.

**Propagator:** Conscious Electromagnetic Information (CEMI) field (McFadden, 2002a, 2002b): the macroscopic electromagnetic field generated by synchronised neural firing across the cortex. Measurable by magnetoencephalography (MEG) and magnetocardiography (MCG) at the body surface. Field feeds back onto neuronal firing thresholds (ephaptic gain), producing a self-modulating propagator.

**Mind matrix:** Subjective awareness and associative memory. The Hopfield energy landscape maps traumatic configurations to deep attractor basins; the FM-HN architecture (§7) provides the runtime coupling to the limbic field.

**Consciousness threshold:** Awareness emerges when the CEMI field amplitude  $\phi \geq T_c = \sqrt{2}$  (normalised units). Proved: `UniversalSomaticField.consciousness_dichotomy`.

---

### Scale 7 – Organism ( $10^0$ m)

**Equation parameters:**  $k = \omega/c_{\text{tissue}}$  ( $c_{\text{tissue}}$ : speed of elastic waves in fascia and soft tissue);  $N \approx 10^{14}$ – $10^{15}$ .

**Physical substrate:** The body as a biotensegrity structure (Ingber 1998): a pre-stressed, globally coupled elastic network of skeleton, fascia, muscle, and viscera. Not a stack of parts but a continuous wave-bearing medium.

**Propagator:** Full somatic CEMI field (the complete 11D configuration space of equation (5)). The cardiac electromagnetic field — the loudest electromagnetic event the body produces — is detectable by MCG at distances up to 2 m from the body surface.

**Mind matrix:** The full 11D organism; all four subspaces active. Subjective experience, emotional regulation, trauma, creativity. The FM-HN architecture (§7) governs runtime dynamics.

---

### Scale 8 – Animal Swarms ( $10^0$ – $10^1$ m)

**Equation parameters:**  $k \sim r_{\text{align}}^{-1}$  (alignment radius);  $N$  = swarm size.

**Physical substrate:** Discrete agents (birds, fish, insects, drones) in 3-dimensional space, each responding to local neighbours.

**Propagator:** Active-matter velocity field (Toner and Tu 1995):  $\partial_t \mathbf{v} + \lambda(\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla P + D_T \nabla^2 \mathbf{v}$ . Global formation emerges from local interactions propagated through the swarm by the same Green's function structure as all preceding scales.

**Key result (swarm coordination, P16 (Johnson, 2026b)):** Treating the swarm as a macroscopic brane projection reduces coordination cost from  $O(N \cdot K)$  to  $O(N^2)$  with  $K = 1$ . The Green's function replaces  $K$  rounds of message-passing with a single matrix-vector product. Jam resistance follows as a corollary ( $K = 1$  means no communication round to disrupt). Proved: `SwarmPropagator.propagator_beats_classical`, `SwarmPropagator.jam_resistant`.

---

### Scale 9 – Society / City ( $10^3$ m)

**Equation parameters:**  $k = r_{\text{interaction}}^{-1} \approx 10^{-3} \text{ m}^{-1}$ ;  $N \approx 10^6\text{--}10^7$  (city population).

**Physical substrate:** Urban infrastructure; transport networks; population distribution. The city as a physical coupling network.

**Propagator:** Social interaction kernel  $G_{ij}$ —how frequently agent  $i$  encounters agent  $j$  in the physical medium. Cultural propagation (dialect spread, technology adoption) is a structural contagion wave governed by the social Green's function:  $P(s_i \rightarrow 1) = \sigma(\sum_j G_{ij}s_j - \theta)$  (social Hopfield network).

**Mind matrix:** Cultural attractors ( $N \approx 10^3$  stable cultural modes). Language variants, norms, and fashions are attractor states of the social field. Estuary English propagation along the Thames Valley is a worked example of geographic boundary conditions selecting which modes propagate and which decay (§12).

### Scale 10 – Geological ( $10^5$ m)

**Equation parameters:**  $k = \omega/v_P \approx \omega/(6000 \text{ m/s})$  (P-wave velocity); boundary: crustal moho below, free surface above.

**Physical substrate:** Tectonic plates; the Alpine fold-and-thrust belt; the Klöntalersee basin (Glarus, Switzerland) as a natural acoustic resonator. The Glarus Hauptüberschiebung places Verrucano sandstone (250 Ma) over Eocene flysch (35 Ma), recording 35 km of northward transport — a wave with a ten-million-year period.

**Propagator:** Seismic Green's function, measurable by global seismometer networks. The Earth's normal modes (free oscillations) are eigenstates of the elastic Green's function under spherical boundary conditions.

**Mind matrix:** Crustal stress distribution tensor ( $N \approx 10^3$  stress modes). Geological memory: the fold geometry of a mountain range encodes every collision the lithosphere has experienced. The rock face is a four-dimensional document read as a three-dimensional spatial slice.

### Scale 11 – Planetary ( $10^6$ m)

**Equation parameters:** Navier-Stokes + heat equation in a rotating frame; effective  $k$  set by thermodynamic convection wavelengths.

**Physical substrate:** Planetary mantle and core. The mantle convects on timescales of millions of years under gravitational and thermal forcing. The geodynamo (liquid outer core) generates the planetary magnetic field.

**Propagator:** Thermodynamic convection kernel; seismic tomography provides the empirical Green's function for the Earth's interior.

**Mind matrix:** Global energetic equilibrium; carbon cycle; ice-age attractor sequence. The climate system is a dynamical system with at least two stable attractors (glacial and interglacial) separated by a bifurcation controlled by orbital forcing (Milankovitch cycles).

---

## Scale 12 – Orbital ( $10^9$ m)

**Equation parameters:** Newtonian gravity;  $k \rightarrow 0$  (long-range, massless graviton).

**Physical substrate:** Planetary and lunar orbits; the heliosphere; Lagrange points. Solar wind creates an effective medium with frequency-dependent propagation properties.

**Propagator:** Gravitational Coulomb kernel  $G_{\text{grav}}(r) = -Gm/r$  (same  $1/r$  form as the electromagnetic Coulomb kernel of Scale 3, with a different coupling constant and sign). Gravitational lensing provides a direct measurement of  $G_{\text{grav}}$ .

**Mind matrix:** Orbital resonance structure. The solar system's Kirkwood gaps and mean-motion resonances are the stable eigenstates of the gravitational  $N$ -body problem.

---

## Scale 13 – Stellar ( $10^{11}$ m)

**Equation parameters:**  $k = \omega/c_s$  (sound speed in stellar plasma  $c_s \approx 100$  km/s);  $N \approx 10^6$  oscillation modes.

**Physical substrate:** Stars: thermonuclear plasma in hydrostatic equilibrium, bounded by radiation pressure and gravity.

**Propagator:** Helioseismic Green's function, directly measured by the Solar Dynamics Observatory. The Sun supports approximately  $10^7$  simultaneous acoustic modes (p-modes, g-modes, f-modes) spanning 5 minutes to hours.

**Mind matrix:** Stellar oscillation spectrum. The eigenfrequency distribution encodes the interior structure — density profile, rotation, chemical stratification. Asteroseismology reads the mind matrix of distant stars.

---



## Scale 14 – Black Holes and Compact Objects ( $10^3$ – $10^{10}$ m)

**Equation parameters:**  $k = 2\pi f_{\text{ISCO}}/c$  (innermost stable circular orbit frequency); boundary: event horizon.

**Physical substrate:** Neutron stars; black holes; binary inspiral systems. The extreme curvature regime of general relativity.

**Propagator:** Quasi-normal modes — the gravitational wave propagator for a perturbed black hole. Each quasi-normal mode is a damped oscillation:  $G_{\text{BH}}(t) \propto e^{-t/\tau_{\text{ring}}} \cos(\omega_{\text{QNM}} t)$ . This is the impulse response of curved spacetime.

**Mind matrix:** Black hole thermodynamic state (Bekenstein entropy  $S = A/4G\hbar$ , where  $A$  is the horizon area). The Bekenstein-Hawking entropy encodes  $\sim 10^{77}$  bits for a solar-mass black hole.

---

## Scale 15–16 – Galactic ( $10^{20}$ – $10^{22}$ m)

**Equation parameters:** Poisson-Vlasov system;  $k \sim \pi/R_{\text{arm}}$  (spiral arm half-wavelength).

**Physical substrate:** Stellar populations ( $\sim 10^{11}$  stars per galaxy); dark matter halo; interstellar medium.

**Propagator:** Density-wave kernel (Lin and Shu 1964). Spiral arms are not physical structures of permanently bound stars; they are density waves — compressions propagating through the stellar fluid. The Green's function of the galactic disk determines which pattern speeds are stable.

**Mind matrix:** Galactic kinematics; rotation curve; spiral arm pattern.  $N \approx 10^{11}$  (number of stars in the Milky Way). The rotation curve encodes the mass distribution including dark matter.

---

## Scale 17–18 – Large-Scale Structure ( $10^{23}$ – $10^{24}$ m)

**Equation parameters:** Linearised cosmological perturbation theory;  $k \sim k_{\text{BAO}} = 0.1 \text{ Mpc}^{-1}$  (baryon acoustic oscillation scale).

**Physical substrate:** Galaxy clusters; cosmic filaments; voids. The large-scale structure traces the initial density perturbations from inflation, propagated through the baryon-photon fluid before recombination.

**Propagator:** Baryon Acoustic Oscillation (BAO) kernel: the Green's function of acoustic waves in the primordial plasma, imprinted on the matter distribution at recombination. BAO provides a standard ruler for cosmological distance measurement.

**Mind matrix:** Large-scale structure topology; cosmic web connectivity.  $N \approx 10^{14}$  (number of galaxies in the observable universe).

---

## Scale 19–20 – Observable Universe ( $10^{26}$ m)

**Equation parameters:** Linearised Einstein equation  $\square h_{\mu\nu} = -16\pi G T_{\mu\nu}$ ;  $k = \omega/c$ ;  $N \rightarrow \infty$ .

**Physical substrate:** The full observable universe from the surface of last scattering ( $z = 1100$ ) to the Hubble sphere ( $c/H_0 \approx 4.4$  Gpc). Dark energy dominates the current energy budget.

**Propagator:** Gravitational wave propagator — the retarded Green’s function of the linearised Einstein equation:  $G_{\text{GW}}(x, x') = \theta(t - t')\delta((x - x')^2)/(2\pi)$ . This is spacetime’s impulse response. Gravity is the Green’s function of the metric field. LIGO/Virgo/KAGRA measure  $G_{\text{GW}}$  directly.

**Mind matrix:** The global cosmological state. If the universal CEMI field amplitude satisfies  $\phi_{\text{cosmic}} \geq T_c$ , the universe satisfies the structural requirements for consciousness (`UniversalSomaticField.universe_is_11D_organism`, axiom). Whether this condition is met dynamically is an empirical question.

**Same as always.** The gravitational wave propagator at Scale 20 is formally identical to the Coulomb propagator at Scale 3 (both are  $1/r$  forms of equation (1) with  $k = 0$ ) and to the synaptic transfer function at Scale 5. One equation. Twenty scales.

---

## Consciousness as Phase Transition

### 6.1 Definition

**Definition (Pre-conscious state).** A system at Scale 6–7 is *pre-conscious* when its limbic field amplitude  $\phi < T_c$ . Field propagation occurs; no first-person awareness is present.

**Definition (Conscious state).** A system is *conscious* when  $\phi \geq T_c$ . The limbic field couples the somatic and cortical subspaces; first-person awareness emerges as a property of this coupling.

**Theorem (Lean 4 verified, `UniversalSomaticField.consciousness_dichotomy`):** For any  $\phi \in \mathbb{R}$ , either  $\phi < T_c$  (pre-conscious) or  $\phi \geq T_c$  (conscious). The transition is sharp.  $\square$

**Theorem (Lean 4 verified, `UniversalSomaticField.consciousness_monotone`):** Raising  $\phi$  cannot destroy consciousness.  $\square$

### 6.2 The Hard Problem

The “hard problem of consciousness” (Chalmers 1995) asks why physical processes give rise to subjective experience. On the present account, the question is reframed: *why does the limbic field amplitude exceed  $T_c$ ?* This is an empirical question about field dynamics, not a philosophical puzzle about the relation between matter and mind.

WKB tunnelling amplitude  $\Theta(W)$  vs. barrier height  $W$ . QUANT-EXP-1 values ( $W = 8, 10, 12$ ) marked. Classical rate = 0; quantum rate =  $\Theta > 0$  always. (Generated: *FS6\_wkb\_amplitude.png*)

Figure 7: WKB tunnelling amplitude  $\Theta(W)$  vs. barrier height  $W$ . QUANT-EXP-1 values ( $W = 8, 10, 12$ ) marked. Classical rate = 0; quantum rate =  $\Theta > 0$  always. (Generated: *FS6\_wkb\_amplitude.png*)

A conscious percept is a **pole in the propagator**: the field’s first-person experience of its own impulse response, occurring when the excitation frequency matches a natural resonance of the manifold. The “felt quality” (quale) is the resonance; the “content” is the mode structure.

### 6.3 The Trauma Attractor

Trauma is a topological obstruction: a configuration of the limbic field  $L_1$  with a high-barrier double well. Classical gradient descent cannot escape:

$$\frac{dx}{dt} = -V'(x) = -4Wx(x^2 - 1) < 0 \quad \text{for } x \in (-1, 0)$$

This traps the system near  $x = -1$  indefinitely. Quantum tunnelling provides the only escape route. The WKB amplitude:

$$\Theta(W) = \exp\left(-\frac{8\sqrt{2W}}{3}\right) \quad (12)$$

is strictly positive for all finite  $W$  (proved: *LimbicTunnel.wkbAmplitude\_pos*) but exponentially small for large  $W$ . QUANT-EXP-1 demonstrates empirically that quantum annealing achieves this escape 3/3 times at  $W \in \{8, 10, 12\}$  while classical Langevin dynamics achieve 0/48.

## The Field-Modulated Hopfield Network

### 7.1 Architecture

The FM-HN ((Johnson, 2026c)) unifies the classical 1982 Hopfield network (Hopfield, 1982) and the modern 2020 network (Ramsauer et al., 2020) as limiting cases of a single architecture parameterised by the limbic field amplitude  $\Phi$ .

Two runtime coupling equations:

$$T(t) = T_0 + \sigma \cdot \Phi_{\text{limbic}}(t) \quad (13)$$

$$W(t) = W_0 + \gamma \cdot \Phi_{\text{limbic}}(t) \cdot J \quad (14)$$

where  $T_0 > 0$  is the baseline temperature,  $\sigma > 0$  the limbic coupling strength,  $J \in \mathbb{R}^{D \times D}$  the coupling matrix, and  $\gamma > 0$  the ephaptic gain coefficient.

## 7.2 Correspondence Principle

**Theorem (Lean 4 verified, `LimbicHopfield.correspondence_principle`):** Under zero somatic stress  $\Phi = 0$ :

$$T(t) = T_0, \quad W(t) = W_0 \quad (15)$$

Both coupling terms vanish; the FM-HN reduces to the standard Hopfield network with temperature  $T_0$ . As  $T_0 \rightarrow 0$  ( $\beta \rightarrow \infty$ ):

$$\text{FM-HN update} \rightarrow \text{sign}(W_0 \cdot s) = \text{HN-1982 update}$$

Proof: `calm_temp_is_baseline` and `calm_weight_is_baseline` by simp.  $\square$

This is the Einstein-Newton relationship for neural architectures: the 1982 network is the low-temperature, calm-somatic limit of the FM-HN, just as Newtonian mechanics is the low-velocity limit of special relativity.

## 7.3 Neurodivergent Operator Modifications

The FM-HN parameter space  $(\beta, W)$  contains distinct regimes corresponding to neurodivergent profiles (all proved by `linarith` in `LimbicHopfield`):

| Profile | Baseline $T$           | Barrier $W$       | Dynamical regime                          |
|---------|------------------------|-------------------|-------------------------------------------|
| ADHD    | $1.8 \cdot T_0$ (hot)  | Low               | Rapid exploration, low settling           |
| ASC     | $0.4 \cdot T_0$ (cold) | Normal            | Deep attractors, rare transitions         |
| C-PTSD  | $T_0$                  | High ( $W = 12$ ) | Classical trapping, quantum escape needed |

**Theorem (Lean 4 verified, `LimbicHopfield.adhd_hotter_than_autism`):**  $T_{\text{ASC}} < T_0 < T_{\text{ADHD}}$ .  $\square$

The Arnold tongue: stable frequency-locked region (shaded) in the parameter space of coupling strength vs. frequency detuning. Rapport = operating inside the tongue. (Generated: *FS8\_arnold\_tongue.png*)

Figure 8: The Arnold tongue: stable frequency-locked region (shaded) in the parameter space of coupling strength vs. frequency detuning. Rapport = operating inside the tongue. (Generated: *FS8\_arnold\_tongue.png*)

## The Relational Field

When two organisms interact, the 11D decomposition extends to a coupled system. The single-organism propagator  $G \in \mathbb{R}^{N \times N}$  becomes a block matrix:

$$\mathbf{G}_{AB}(\omega) = \begin{pmatrix} G_{AA}(\omega) & G_{AB}(\omega) \\ G_{BA}(\omega) & G_{BB}(\omega) \end{pmatrix} \quad (16)$$

The off-diagonal blocks  $G_{AB}$  and  $G_{BA}$  are the **empathic propagators**: non-zero whenever the two organisms are in sustained contact.

**Huygens frequency locking.** Two coupled oscillators with natural frequencies  $\omega_A$  and  $\omega_B$  and coupling  $\kappa = |G_{AB}|$  lock to a common frequency when:

$$|\omega_A - \omega_B| < \kappa \quad (17)$$

(Arnold tongue condition). Rapport is the phenomenological signature of frequency locking. The Arnold tongue width grows with friendship depth (persistent  $G_{AB}$ ), explaining why close relationships synchronise across longer frequency separations.

**Therapist-client entrainment.** The therapist's regulated field (low  $W_T$ , stable attractor) modifies the client's effective barrier via:

$$W_{\text{eff}} = W \cdot (1 - \alpha |G_{TC}|^2) \quad (18)$$

As therapeutic alliance deepens ( $|G_{TC}|^2$  grows),  $W_{\text{eff}}$  decreases and the WKB tunnelling amplitude  $\Theta(W_{\text{eff}})$  increases. This provides a quantitative prediction: the depth of therapeutic alliance should predict the rate of symptomatic improvement in trauma-spectrum conditions.

## Encapsulation of Related Frameworks

The zUSF encapsulates three existing frameworks as special cases or scale-restricted projections.

### 9.1 McFadden's CEMI Theory (McFadden, 2002a, 2002b)

McFadden proposes that consciousness correlates with the brain's endogenous electromagnetic field. In the zUSF: the CEMI field is the Scale-6 ( $\sigma = 6$ ) restriction of the universal propagator  $G$ . The zUSF extends CEMI in two directions: downward to quantum neural noise (Scale 5) and upward to multi-organism coupling (§8) and cosmological propagation (Scale 20).

### 9.2 Schreiber's Modal Homotopy Type Theory

Schreiber (2013) formalises physics in dependent type theory, arriving at an 11-dimensional structure from the mathematics of M-theory. The zUSF arrives at the same 11-dimensional structure from the bottom up (clinical observation). The structural isomorphism (theorem 3.2) confirms that the two approaches describe the same object. The zUSF provides the biological execution engine that Schreiber's purely mathematical framework lacks.

### 9.3 Hoffman's Conscious Agents Model (Hoffman, 2019)

Hoffman proposes that spacetime is a “user interface” constructed by conscious agents; it is not fundamental. The zUSF disagrees on one point: spacetime ( $D_{1-4}$ ) is physically real and causally efficacious. Brain surgery alters subjective experience because physical processes in spacetime causally affect the CEMI field. However, the zUSF agrees that the deeper structure is relational: conscious percepts are poles in the propagator — relational objects, not substances. The “conscious agents” in Hoffman's framework correspond to 11D organisms that have crossed the threshold  $T_c$ .

## Formal Verification

The core algebraic results are Lean 4 kernel-verified using Mathlib (v4.28.0). The following table lists theorems, proof methods, and files.

| Theorem                            | Statement                                         | Tactic              | File                            |
|------------------------------------|---------------------------------------------------|---------------------|---------------------------------|
| <code>dim_is_11</code>             | $4 + 3 + 1 + 3 = 11$                              | <code>decide</code> | <code>MTheoryIsomorphism</code> |
| <code>somaField_iso_mtheory</code> | $\text{SomaField} \cong \text{M-Theory}$          | <code>simp</code>   | <code>MTheoryIsomorphism</code> |
| <code>organism_hierarchy</code>    | $11D \twoheadrightarrow 7D \twoheadrightarrow 4D$ | <code>simp</code>   | <code>MTheoryIsomorphism</code> |
| <code>scale_iso_commutates</code>  | $\Lambda$ commutes with scale transform           | <code>simp</code>   | <code>MTheoryIsomorphism</code> |

| Theorem                    | Statement                                | Tactic              | File                  |
|----------------------------|------------------------------------------|---------------------|-----------------------|
| boundary_not_interior      | $L_1$ endpoints $\notin$ interior        | fin_cases           | MTheoryIsomorphism    |
| V_nonneg                   | $V(x) \geq 0$ everywhere                 | positivity          | LimbicTunnel          |
| barrier_height             | $V(0) = W$                               | simp, ring          | LimbicTunnel          |
| gradient_traps_near_neg    | Classical trapped near $x = -1$          | nlinarith           | LimbicTunnel          |
| wkbAmplitude_pos           | $\Theta(W) > 0$ for all $W$              | exp_pos             | LimbicTunnel          |
| wkbAmplitude_lt_one        | $\Theta(W) < 1$ for $W > 0$              | analysis            | LimbicTunnel          |
| correspondence_principle   | EM-HN = standard HN at $\Phi = 0$        | simp                | LimbicHopfield        |
| stress_raises_temp         | $\Phi > 0 \Rightarrow T > T_0$           | linarith            | LimbicHopfield        |
| adhd_hotter_than_autism    | $T_{\text{ASC}} < T_0 < T_{\text{ADHD}}$ | linarith            | LimbicHopfield        |
| propagator_beats_classical | $N^2 < NK$ for $K > N$                   | Nat.mul_lt_mul_left | SwarmPropagator       |
| jam_resistant              | Propagator: $K = 1$                      | rfl                 | SwarmPropagator       |
| consciousness_dichotomy    | $\phi < T_c$ or $\phi \geq T_c$          | lt_or_le            | UniversalSomaticField |
| consciousness_monotone     | Raising $\phi$ preserves consciousness   | linarith            | UniversalSomaticField |
| universal_field_theory     | $G$ is scale-invariant                   | structural          | UniversalSomaticField |

#### Axioms (not yet proved; explicit gaps):

| Axiom                       | Content                               | Scaffolding needed               |
|-----------------------------|---------------------------------------|----------------------------------|
| greens_fn_is_SHO            | $G$ satisfies SHO equation            | Schwartz distribution theory     |
| universe_is_11D_organism    | Universe satisfies 11D structure      | Cosmological boundary conditions |
| cosmological_correspondence | Scale 19 instantiates equation (1)    | Linearised GR in Mathlib         |
| classical_trapped           | Gradient flow stays in $(-\infty, 0)$ | Lyapunov theory for ODEs         |
| quant_exp_1_formal          | Quantum rate $>$ classical rate       | Probabilistic model of annealing |

Every result not on the axiom list is kernel-verified. No sorry. No admit.

## Falsifiability and Predictions

### 11.1 Testable predictions

1. **Therapeutic alliance and barrier height.** Equation (18) predicts that  $W_{\text{eff}}$  decreases linearly with  $|G_{TC}|^2$ . Measuring the Working Alliance Inventory (WAI) score as a proxy for  $|G_{TC}|^2$  and PTSD symptom severity as a proxy for  $W_{\text{eff}}$ , the model predicts a significant negative correlation between WAI and symptom reduction rate, even after controlling for treatment modality.
2. **Neurodivergent temperature profiles.** The model predicts that ADHD individuals should show elevated resting-state neural temperature (higher effective  $\beta^{-1}$  in attractor dwell-time distributions) relative to ASC individuals in a same-task fMRI paradigm.
3. **Swarm coordination speedup.** The  $O(N^2)$  propagator protocol should outperform  $O(N \cdot K)$  message-passing for  $K > N$ . This is directly testable with autonomous vehicle fleets ( $N = 100$ ,  $K$  measured empirically).
4. **WKB barrier sweep.** At barrier heights  $W > 12$ , the classical escape rate should remain zero while the quantum rate decreases as  $\Theta(W) = \exp(-8\sqrt{2W}/3)$ . This prediction is testable on D-Wave hardware by extending the QUANT-EXP-1 protocol to  $W \in \{14, 16, 18\}$ .

### 11.2 Falsification conditions

The framework is falsified if any of the following is observed:

- Classical Langevin dynamics escape the barrier in QUANT-EXP-1 at rate  $> 0$  (contradicts `LimbicTunnel.classical_trapped`)
- The FM-HN under zero stress produces different output than the classical 1982 network (contradicts `correspondence_principle`)
- The propagator coordination protocol fails to reduce to  $O(1)$  application steps (contradicts `jam_resistant`)
- Two systems with type-mismatched scale parameters successfully couple (contradicts the dependent-type architecture of the Zoom Operator)

## Discussion

### 12.1 Scope and Limitations

The zUSF is a structural claim. It asserts that the same equation governs propagation at all scales; it does not assert that all scales are phenomenologically equivalent or that cosmological structures are conscious in the same sense as biological organisms. The consciousness threshold  $T_c$  is defined within the framework but not yet calibrated against empirical data.



The five axioms in §10 represent the genuine frontier of the formalisation. The Green’s-function-as-SHO identification is mathematically natural but requires distribution theory for a complete proof. The cosmological claims require linearised general relativity in Mathlib.

## 12.2 The Inductive vs. Deductive Derivation

Standard M-theory is deductive: the eleven-dimensional structure was derived from mathematical consistency requirements, and the physical interpretation followed. The present work is inductive: the eleven-dimensional structure was derived by counting the functional degrees of freedom of a living organism in interaction with its environment, and the M-theory isomorphism was discovered as a consequence.

This matters epistemologically. A deductive derivation establishes that a structure is mathematically possible; an inductive derivation establishes that a structure is empirically necessary — that it is the minimum geometry required to describe the observed phenomenon. The zUSF claims necessity, not merely possibility.

## 12.3 Relation to Existing Work

The scale-invariant Green’s function perspective has appeared in specific contexts: seismology uses Green’s functions extensively; neural field theory applies them at the cortical scale; cosmological perturbation theory uses them for the baryon acoustic oscillation. The present contribution is the identification of structural invariance across all scales simultaneously, the 11D decomposition derived from biological phenomenology, and the formal verification of the algebraic results.

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## Open Research Problems

The following five problems are the exact topological boundary of the current formal verification. Everything not on this list is proved. These are not vague limitations — each has a known mathematical target and a clear path to closure.

**Problem 1: The Green’s Function SHO Identity (distribution theory).** The axiom `greens_fn_is_SHO` in `UniversalSomaticField.lean` states that the Green’s function  $G(x, x')$  satisfies the SHO equation  $(\partial_{x'}^2 + k^2)G(x, \cdot) = \delta(\cdot - x)$  in the sense of distributions. The proof requires Schwartz space and tempered distribution theory. Mathlib has `MeasureTheory` and `Distribution`-adjacent infrastructure; the specific result (fundamental solution of  $-\nabla^2 + k^2$ ) is not yet in Mathlib as a verified theorem. **Path to closure:** contribute the Yukawa/Helmholtz Green’s function to Mathlib, then discharge the axiom.

**Problem 2: The  $G_2$  Compactification Derivation.** The 7 compact dimensions are currently *postulated* to correspond to the BRECVEMA mechanisms. A complete derivation would proceed from a neurodynamical Lagrangian  $\mathcal{L}[\psi, \partial\psi]$  over an 8D state space, vary it to obtain the Euler–Lagrange

equations, and show that the resulting moduli space has the homotopy type of a  $G_2$ -holonomy manifold. This would replace an identification with a derivation. **Path to closure:** construct the variational problem over the BRECVEMA space; use Mathlib’s `VariationalCalculus` when available.

**Problem 3: The `FieldLayerType` Functor Upgrade.** The `FieldLayerType` encoding in `MTheoryIsomorphism.lean` uses `String` placeholders for the physical content of each layer (`"NavierStokesFlow"`, `"EinsteinGR"`, `"HopfieldHamiltonian"`). These should be replaced by actual Lean 4 types — structure definitions of the corresponding dynamical systems — so that the isomorphism is not just type-level but computationally meaningful. **Path to closure:** define `NavierStokesField`, `EinsteinMetric`, `HopfieldNet` as Lean structures; replace the `String` tags with these types.

**Problem 4: Path-Dependence in Moduli Space.** The dissonance coordinate in `manifold_coords.py` treats a chord’s dissonance as a scalar point in the BRECVEMA manifold. Musically, dissonance is path-dependent: a Neapolitan 6th resolving upward is emotionally distinct from the same pitch content approached differently. The correct formalisation uses a path  $\gamma : [0, 1] \rightarrow \mathcal{M}$  through the  $G_2$  moduli space, with the monodromy of the holonomy connection recording the path-history. **Path to closure:** extend `GeographicSomatic.lean` (once written) to use `PathIntegral` machinery; update `manifold_coords.py` accordingly.

**Problem 5: The Dyadic Coupling Inequality (Float arithmetic).** `DyadicField.lean` contains one sorry: the theorem that dyadic coupling lowers energy when  $J \geq 0$  and both fields have non-negative activation. The proof is straightforward over  $\mathbb{R}$  (the cross-coupling sum  $\sum_{i,j} a_i J_{ij} b_j \geq 0$  when  $a_i, b_j, J_{ij} \geq 0$ ), but Lean 4’s `Float` type is axiomatized and not amenable to algebraic tactics (`linarith`, `nlinarith` do not apply to `Float`). **Path to closure:** re-implement the key energy functions over  $\mathbb{R}$  using Mathlib’s `Real` type; the `Float` implementations can remain as computational code while the proofs use the `Real`-valued versions. This is a refactoring task, not a mathematical problem.

## Conclusion

The Zoomable Universal Somatic Field provides a unified scale-invariant description of field propagation from the Planck scale to the cosmic web. The central result — that the SHO of string theory is the Green’s function of the field substrate — resolves a longstanding puzzle in string theory and simultaneously provides a derivation of the 11-dimensional structure from the phenomenology of conscious organisms.

The architecture satisfies three independent criteria for a successful unification theory:

1. **Structural necessity.** The 11-dimensional decomposition is not a convenient choice; it is the minimum number of functional degrees of freedom required to describe a living system with body, field, homeostasis, and mind.

2. **Formal verification.** The core algebraic results are machine-checked. The axiom list is explicit and minimal.
3. **Falsifiability.** The framework makes specific quantitative predictions (§11.1) that distinguish it from its competitors.

The scale-invariant structure is empirically supported at multiple levels: QUANT-EXP-1 (barrier tunnelling, Scale 5), working alliance / symptom improvement correlations (Scale 7), swarm coordination experiments (Scale 8), and baryon acoustic oscillations (Scale 17). The remaining predictions (§11.1 items 2–4) are testable with currently available hardware.

The equation is simple. The implications are large.

$$(\nabla^2 + k^2) G(x, x') = \delta(x - x')$$


---

## Introduction

Geophysics has two problems that remain open despite decades of effort: earthquake prediction and the mechanism of fault criticality. The first is famously hard; the second is theoretically well-characterised but poorly connected to observational signatures.

The Universal Somatic Field provides a new perspective. At the geological scale, the Earth's crust is a medium with a characteristic wave velocity — exactly the situation described by the Helmholtz master equation. Seismic waves are the Green's functions of this medium. Tectonic stress accumulation is a field-level analogue of the trauma topology described in the clinical USF.

The identification is exact. The consequences are testable.

## The Master Equation at the Geological Scale

At scale 10–12 (crustal thickness to tectonic plate scale), the Helmholtz equation governs seismic wave propagation:

$$(\nabla^2 + k_{\text{seis}}^2) G_{\text{seis}}(x, x') = \delta(x - x')$$

where  $k_{\text{seis}} = \omega/v_p$  is determined by the P-wave velocity  $v_p$  of the crustal medium and the frequency  $\omega$ . The Green's function  $G_{\text{seis}}$  is the seismic wave propagator — the response at seismograph location  $x$  to a unit stress perturbation at source  $x'$ .

This is the same master equation as at every other scale in the USF. The wavenumber changes; the form of the equation does not.

**Scale invariance at the geological scale:** The seismic propagator satisfies the same structural theorems as the neural impulse propagator at scale 7 and the gravitational wave propagator at scale 19. This is not an analogy; it is a consequence of scale invariance.

## Tectonic Stress as Somatic Field Energy

Tectonic stress accumulation maps directly to the trauma topology of the clinical USF. A locked fault stores elastic strain energy in a configuration that resists change — this is the geological analogue of the trauma well: a state maintained by topological protection, not energetic preference.

Formally: the stress tensor field  $\sigma_{ij}(x)$  at a locked fault is in a metastable minimum of the Hopfield energy:

$$H_{\text{tect}}[\sigma] = -\frac{1}{2} \int \sigma_{ij}(x) W_{ijkl}(x - x') \sigma_{kl}(x') dx dx'$$

where  $W_{ijkl}$  is the elastic compliance tensor of the crust. The metastable minimum is maintained by the frictional coupling — the geological analogue of the limbic constraint.

An earthquake occurs when this metastable minimum becomes unstable: the spectral gap of the stress tensor closes, and the field undergoes a topological phase transition to a lower-energy configuration.

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## Earthquakes as Phase Transitions

The Gutenberg-Richter law ( $\log N = a - bM$ , where  $N$  is the number of earthquakes with magnitude  $\geq M$ ) is an empirical power law with no deep physical derivation. The USF provides one.

Power-law statistics are the signature of a system near a critical point — a point where the spectral gap of the governing operator approaches zero. The Earth's crust, maintained near criticality by the continuous input of tectonic stress, is precisely such a system.

The USF predicts that:

1. The magnitude-frequency distribution follows a power law because the crust is near a topological critical point — the geological analogue of the consciousness threshold  $T_c$
2. The  $b$ -value in the Gutenberg-Richter law is related to the spectral gap of the stress tensor:  $b \propto \Delta\lambda/k_B T_{\text{eff}}$  where  $\Delta\lambda$  is the spectral gap and  $T_{\text{eff}}$  is the effective temperature of seismic noise
3. Earthquake triggering at small magnitudes deviates from classical threshold models because quantum-mechanical tunnelling contributions become significant — a WKB correction to the classical nucleation rate

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## The Slow Wave: Rock as Long-Memory Field

The geological soma has a characteristic time scale orders of magnitude longer than the biological soma: tectonic cycles operate on millions of years. Rock strata are the geological equivalent of long-term memory — a record of the field's historical attractor traversals encoded in physical stratigraphy.

The USF interpretation: each sedimentary layer is a time-stamped snapshot of the geological field state. Unconformities (missing strata) are topological defects — periods during which the field was in a non-recording state, the geological analogue of dissociation.

This gives a new interpretation of geological history: stratigraphy is the field's Pensieve — its externalised long-term memory.

## Predictions

**Testable prediction 1:** The WKB correction to earthquake nucleation rates predicts a departure from the Gutenberg-Richter power law at small magnitudes ( $M < 1.5$ ). The correction term is:

$$N(M) \propto M^{-b} \cdot e^{-S_{\text{WKB}}(M)}$$

where  $S_{\text{WKB}}(M)$  is the WKB action for the stress field to tunnel through the frictional barrier. This is measurable with high-resolution seismometer arrays.

**Testable prediction 2:** Pre-seismic electromagnetic anomalies (observed empirically but poorly explained) are the geological equivalent of the CEMI field fluctuations that precede emotional state transitions. Their frequency spectrum should match the seismic propagator's imaginary pole — the resonant frequency of the stress field approaching criticality.

**Testable prediction 3:** The spatial correlation length of aftershock sequences follows the Green's function decay of the seismic propagator, matching the  $e^{-kr}/r$  Yukawa form at scale 10.

## Conclusion

Seismic propagation is a Green's function. Tectonic criticality is a topological phase transition. Rock strata are field memory. This is the identification. The method used to find it is documented in the Mathematical Co-identification paper. That method is now history. The structure stands.

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